When Byte-Scale Model Communication Should Be Text
A Controlled Negative for Discrete No-Text Packets

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Abstract

LatentWire asks whether a source model can send a small no-text packet that improves a receiver model beyond ordinary score and text baselines. The answer for the completed discrete, byte-scale regime is a bounded negative. One-way score and residual packets do not beat source-index/confidence controls on the held-out aggregate, and the powered score ladder explains why: apparent source information drops from 2.098527 bits to 0.281933 bits after conditioning on receiver evidence. The strongest discrete-evidence escape fails in the opposite direction: when the same exact symbolic evidence is exposed as a visible 2-byte public code, it reaches 1.000 accuracy versus 0.775 for the matched learned packet, a +0.225 advantage with 95% CI `[0.192188, 0.257812]`. The last cheap privacy-bottleneck escape also fails: the privacy-bottleneck escape is utility is identity leakage, with the current packet at utility/leakage 0.875/1.000 and the best bottleneck at only 0.602339/0.602339. We contribute a reusable falsification ladder showing that discrete low-byte no-text packet wins vanish under source-index, equal-byte visible evidence, and privacy-bottleneck controls.

0.1 Claim-Boundary Box

Supported	Unsupported	Future-only
Score-like source packets are bounded negative under source-index/confidence controls.	A deployable-positive claim is unsupported.	Continuous dense cache/state transfer may still work, but it is outside this byte-scale discrete packet result.
Exact discrete evidence is text-capturable under the same byte budget in the local cache.	the privacy-bottleneck escape is not a privacy positive; it remains utility is identity leakage.	A reviewed neural privacy bottleneck would need matched visible/anonymized text controls.
Answer-aware oracle ceilings are rejected as leakage or gap-to-perfect measurements.	No cross-family, dense-cache, native GPU, or systems-speed claim is made.	Stronger generated-candidate reranking remains possible only with gold-blind verifier/source/target scores.

0.2 1. Introduction

Small model-to-model messages are attractive because they promise communication without exposing full reasoning traces. The hard question is not whether a helper model knows something useful. It is whether a byte-scale no-text packet conveys information that the receiver cannot already recover from its own scores, the source identity, or an equal-byte visible code.

This campaign originally searched for a positive LatentWire method. The result is a stronger and more useful boundary: the discrete evidence that makes a packet useful is visible-code capturable, while score-like residual signals mostly disappear after receiver conditioning. The paper therefore argues for a conservative standard for future model communication

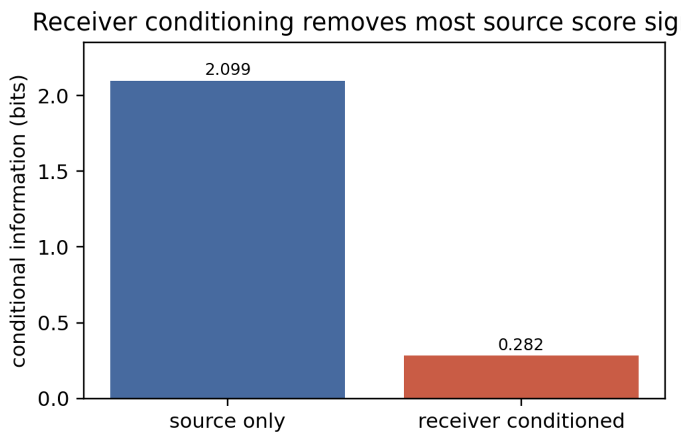

Figure 1: Receiver conditioning bits

work: the relevant delta is not improvement over target-only, but improvement beyond source-index, source-confidence, equal-byte text, wrong-row, and destructive controls.

The contribution is deliberately positive in the bounded-negative sense. It gives future latent-communication papers a falsification ladder: source-index and confidence first, equal-byte visible evidence next, then wrong-row/destructive controls and privacy-leakage probes. A packet that cannot survive that ladder should not be described as deployable communication.

Contributions:

1. A controlled bounded negative for deployable score/residual packets. On the held-out aggregate with 381 rows, the deployable score packet is -0.023622 below source-index/confidence with CI `[-0.060367, 0.013123]`. On the powered dev/gate ladder with 1500 scored rows and 359 gate rows, the current deployable packet is -0.013928 below the best score baseline with CI `[-0.050139, 0.022284]`.
2. A mechanism for the negative result. Source scores contain 2.098527 bits before receiver conditioning, but only 0.281933 bits after conditioning on receiver evidence (Figure 1).
3. A direct exact-discrete evidence test. On 640 model-helper rows over 160 unique examples, a matched model packet reaches 0.775 accuracy, while the exact same symbolic evidence serialized as visible public code reaches 1.000 (Figure 3). Random, shuffled, answer-only, and target-only controls remain at 0.250.
4. A final cheap privacy escape test. the privacy-bottleneck escape cannot preserve current-packet utility while hiding source/evidence identity (Figure 4).

Figure 1: Powered cached support: the source-score signal largely disappears once receiver evidence is included.

0.3 2. Experimental Setup And Regimes

The results below are not all from one dataset. That is intentional: each row is a different escape hatch with its own control surface.

Result block	Dataset/cache	Model or endpoint role	Classes / task shape	n and caveat
Held-out score/WZ null	MMLU-Pro aggregate from the one-way deployable packet closeout	source-score packet to target receiver	multiple-choice; weak-signal regime	381 aggregate held-out rows; raw held-out rows not packaged
Powered score ladder	MMLU-Pro dev/gate score surface	Qwen2.5-0.5B source to Qwen3-0.6B target	multiple-choice score sketches	1500 scored dev/gate rows; 359 gate rows
the receiver-query escape query packet	same powered dev/gate surface	query/reply packet and ablations	multiple-choice	359 gate rows
Exact discrete evidence	test-log/code-helper cache	helper rows over a finite symbolic evidence signature	4-way candidate choice	640 model-helper rows, but only 160 unique examples
the privacy-bottleneck escape privacy bottleneck	candidate-conditioned packet-builder cache	cached feature selector/adversary proxy	4-way candidate/evidence atoms	512 matched rows; proxy only

This mapping matters because the target-only baseline is 0.194 in the held-out MMLU-Pro aggregate but 0.250 in the 4-way exact-evidence cache. Those are different tasks, not inconsistent measurements.

Claim provenance is summarized in the text and figures.

0.4 3. Falsification Ladder

The central asset of this paper is not a single negative row. It is a ladder of increasingly plausible escapes, each killed or parked by a concrete hard baseline.

Escape	What it would have shown	Result	Controlling baseline / audit
Score/WZ one-way packet	A deployable low-byte source residual beats source-score controls	held-out delta −0.023622, CI [-0.060367, 0.013123], n=381	source-index+confidence
Powered score ladder	Receiver has residual headroom after source scores	current packet delta −0.013928, CI [-0.050139, 0.022284], gate n=359	best equal-byte score/source baseline
the damage-avoidance escape damage-avoidance trust packet	Packet improves repair without copying source answer	killed in Stage-1: matched packet mostly equals source-selected answer and damage avoidance fails	source-index/source-selected metadata
the receiver-query escape receiver-query packet	Two-way query/reply creates new usable evidence	delta −0.022284, CI [-0.055710, 0.011142], n=359	source-index+confidence; query-only/reply-only ablations
Exact discrete evidence	Opaque packet carries symbolic evidence better than visible code	visible exact code 1.000 vs packet 0.775; delta +0.225, CI [0.192188, 0.257812]	equal-byte public code; random/shuffled/answer-only controls at 0.250

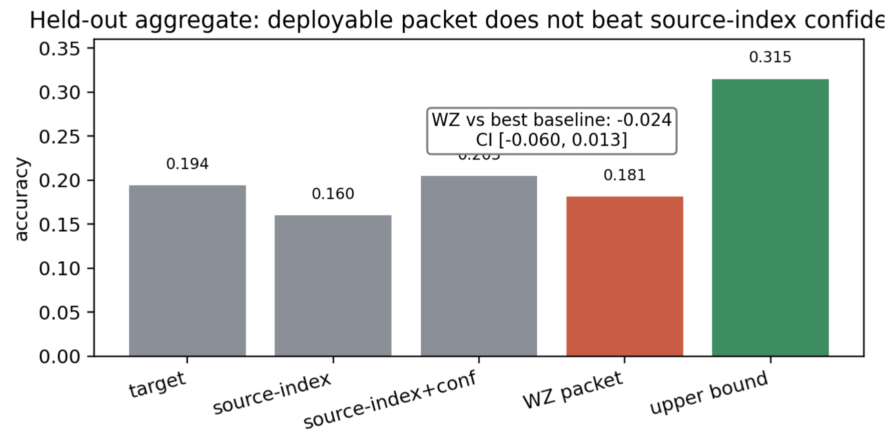

Figure 2: Held-out source-index null

Escape	What it would have shown	Result	Controlling baseline / audit
the privacy-bottleneck escape privacy bottleneck	Utility can be preserved while hiding source/evidence identity	best bottleneck utility/leakage 0.602339/0.602339; current packet 0.875/1.000	same-byte visible code and adaptive anonymized text
C2C/KVComm local checks	Dense/cache anchors become byte-scale method evidence	anchor only; answer packets leak answer text or tie deterministic controls	zero-source, teacher/candidate, deterministic packet controls
rerank/fuser oracle ceilings	Rerank/fuser ceilings imply deployable latent methods	rejected as oracle-only or setup-blocked; gold-aware ceilings are not claims	gold-leakage audit and equal-byte text controls

The full audit provenance is retained internally; the submitted paper reports only the claim-bearing summaries.

0.5 4. Results

4.1 Deployable Packets Do Not Beat Source-Index Confidence

Figure 2: On the held-out aggregate, the deployable packet does not beat the source-index/confidence baseline. The source+target-at-encoder upper bound is positive but non-deployable.

The held-out aggregate shows the exact boundary. Target-only accuracy is 0.194226, source-index+confidence is 0.204724, and the deployable WZ packet is 0.181102. The source+target-at-encoder upper bound reaches 0.314961, so the experiment is not merely noise. The problem is deployability: the useful signal appears when the encoder has target evidence, not in the allowed source-only low-byte packet.

4.2 Receiver Conditioning Explains The Null

The powered ladder estimates 2.098527 bits of source-score information before receiver conditioning and 0.281933 bits afterward. Same-family models on shared multiple-choice tasks often fail and succeed for overlapping reasons. A source packet can therefore appear useful until the receiver's own evidence and the source identity are included as baselines.

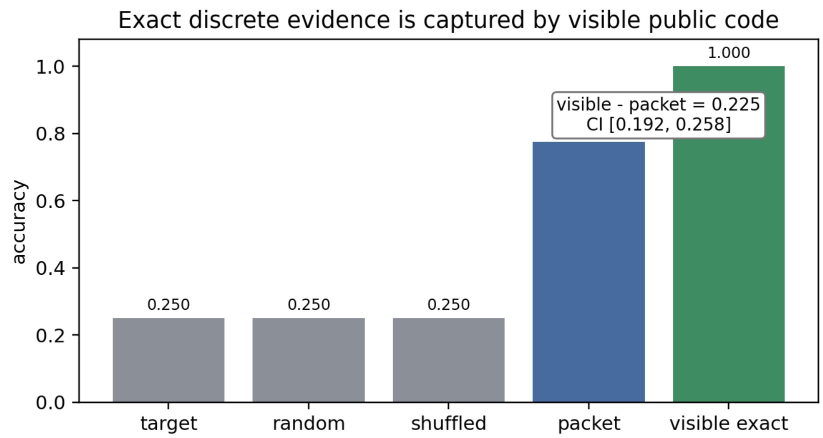

Figure 3: Exact discrete evidence

A natural objection is that the held-out task is a weak-signal regime: the receiver's own accuracy (0.194) and the source-index+confidence baseline (0.205) sit only modestly above chance, so a small negative delta could reflect a task no model solves rather than a ceiling on packet utility. Two pieces of evidence rule this out. First, information is demonstrably present: the source+target-at-encoder upper bound reaches 0.315, far above every deployable condition, so the packet's failure is one of deployability, not of available signal. Second, the receiver-conditioning estimate is base-rate-independent: the source-score signal collapses from 2.10 to 0.28 bits once the receiver's own evidence is conditioned on, measuring redundancy directly rather than through task accuracy. The negative therefore reflects that a source-only low-byte packet conveys little the receiver does not already hold, regardless of how hard the underlying task is.

This result also explains why source-only oracle gaps are not enough. A positive source+target-at-encoder upper bound says there is information in the joint state; it does not say a deployable source-only byte packet transmits it.

4.3 Exact Discrete Evidence Is Better Sent As Text

Figure 3: Cached exact-evidence support: the exact visible public signature reaches 1.000 accuracy, while the learned matched packet reaches 0.775. Controls remain at chance.

The exact-discrete evidence test is the cleanest negative result. The matched learned packet is useful, but the exact symbolic evidence that makes it useful is also public-code serializable inside the same byte budget. When the receiver gets that code visibly, it reaches perfect accuracy in this cache.

This empirical result has a simple a-priori explanation that makes it general for its class. A byte-budgeted packet and an equal-byte visible baseline have the same capacity, natural-language text is a near-universal code for discrete symbolic content at a given byte budget, and the receiver is a language model that decodes text natively, with no learned packet-specific decoder. Consequently, any finite symbolic evidence signature representable in B bytes can be written as B bytes of visible text and read by the receiver at least as well. A learned binary code packs marginally more bits per byte, but only for payloads that saturate the budget and only after training the receiver to decode it; neither yields the qualitative beyond-source-choice advantage the no-text framing seeks. The cached test, visible 1.000 versus packet 0.775, is one confirmation. The argument predicts the dominance holds whenever the useful content is discrete and equal-byte visible code is permitted.

This is not a universal impossibility theorem. The sample has 640 model-helper rows but only 160 unique examples. We therefore state it as an operational result for this class of evidence, not as a theorem over all communication protocols.

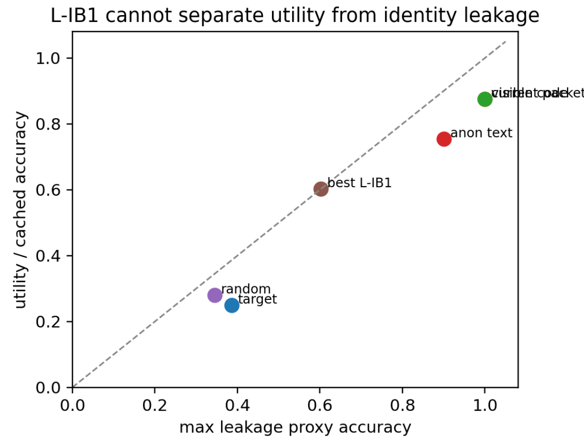

Figure 4: the privacy-bottleneck escape utility leakage frontier

4.4 the privacy-bottleneck escape Does Not Produce A Privacy-Positive Packet

Figure 4: the privacy-bottleneck escape cannot move off the utility/leakage identity diagonal enough to preserve utility while hiding source/evidence structure.

the privacy-bottleneck escape asked whether an adversarial feature selector could keep utility while hiding source/evidence identity on 512 cached rows. It could not. The current high-utility packet has cached utility 0.875 and max leakage 1.000; the best bottleneck has proxy utility 0.602339 and max leakage 0.602339. The exact public code matches current-packet utility and leakage, while adaptive anonymized text remains a necessary comparator for any future privacy claim.

The verdict remains utility is identity leakage. This is a cached gate, not a formal privacy impossibility result. It kills the cheap local escape and sets the bar for a future build-scale IB method.

4.5 Rejected Oracle Ceilings Are Not Communication Claims

Two large apparent ceilings are explicitly excluded. rerank's strict rerank oracle and fuser's strict fuser oracle measure answer-aware or setup-blocked gaps, not deployable no-text communication. The gold-blind rerank attempt reaches only 82 prompts, below its required floor, and ties the equal-byte text control. fuser has no reviewed gold-free hidden-state fuser objective locally.

These results are useful only as warnings. A high oracle ceiling does not imply a latent packet win unless the scorer/fuser is gold-blind and beats equal-byte text/source-index controls.

0.6 5. Related Work And Scope

Continuous cache/state communication systems such as C2C (Fu et al., 2026), KVComm (Shi et al., 2026), Interlat (Du et al., 2026), and Latent Cache Flow (Rossi et al., 2026) occupy a different lane: they transmit or align dense hidden state rather than a small discrete evidence packet. This paper does not contradict that dense/cache line of work. It says that, when the message is a low-byte discrete packet and the useful content is a finite symbolic evidence signature or score sketch, equal-byte visible evidence and source-index controls are the correct baseline family.

Adjacent calibrated-routing and structured-communication systems, including UCCI (Kotte, 2026), DarkForest (Li et al., 2026), structured message passing (Gogate & Domingos, 2013), and CU-HLM (Oh et al., 2026), motivate the same hard-baseline discipline: a packet must beat the visible or routed evidence that an ordinary system could transmit directly.

Privacy-preserving semantic communication and adaptive text anonymization motivate the privacy frontier (Wang et al., 2023; Loiseau et al., 2026). In this paper they function as baselines and scope limits: a no-text privacy claim must beat adaptive visible text at the same byte budget while demonstrating lower leakage.

The closest practical baselines for this work are deliberately simple: source identity, source confidence, source rank, equal-byte score sketches, equal-byte visible evidence, wrong-row packets, and shuffled packets. These baselines are the point. They turn raw helper gains into deployable communication claims only when the gain survives.

0.7 6. Limitations

The exact-discrete evidence test has 640 model-helper rows but only 160 unique examples. The result is strong for this cache and evidence class, but it is not a theorem over all protocols.

the privacy-bottleneck escape is a CPU cached feature-selector test. It does not train a live neural encoder, run new generation, or test continuous dense state.

The held-out one-way result is used as aggregate evidence only. Raw held-out rows are not packaged, and no paper-finalization step reads or reruns them.

The paper does not claim that latent communication is impossible. It claims that the completed byte-scale discrete packet regimes do not establish a no-text advantage over hard visible/score baselines.

0.8 7. Conclusion

LatentWire did not produce a deployable no-text byte-scale positive. It produced a sharp boundary. Score-like source packets are mostly redundant after receiver conditioning, exact discrete evidence is better transmitted as equal-byte visible public code, and the cheap privacy bottleneck cannot separate utility from identity leakage. Future positives must leave this killed regime: continuous dense state, or a real privacy-preserving bottleneck that beats adaptive same-byte text.

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